8°/Hemispherical Reflectance Multi-Step Target Certificate

Rendered To:

Authorization: Sales Order:

Calibration Laboratory: Labsphere, Inc., Reflectance Calibration Laboratory
231 Shaker Street
North Sutton, NH 03260
Tel 603-927-4266 Fax 603-927-4694

Description of Calibrated Items
One white diffuse reflectance sample, model SRT-MS-050 Multi-Step Reflectance Target, serial number MS050-0419-1952

Calibration
8°/Hemispherical (8/h) Spectral Reflectance Calibration over the range 250 – 2500nm, reported at 50nm intervals

Description of Calibration
The calibration items are measured using a Perkin Elmer Lambda 900 or Lambda 950 dual beam spectrophotometer with Labsphere PELA-1000 integrating sphere reflectance accessories which perform the 8/h reflectance measurements. The instrument used for the measurement of the calibrated items is identified with an X in Table I below.

Table I Measurement Instrument and Accessory

Spectrophotometer	Lambda 900A S/N 101N 3060902	Lambda 900B, S/N: 70038	Lambda 950C SN: 950N9031801
Accessory	LambdaX 150mm InGaAs RSA Serial No. 0508182129	LambdaX 150mm InGaAs RSA Serial No. 0508182127	LambdaX 150mm InGaAs RSA Serial No. 0508182128
Instrument Used	X		

The integrating sphere diameter is 150 mm, the sample port diameter is 25 mm, and the interior material is Spectralon. The reference standard and sample piece are placed sequentially in the sample port of the sphere. The reference beam of the instrument acts as an auxiliary beam to correct for substitution error in the sphere. The calibration of the instrument follows the NIST method of utilizing pressed polytetrafluoroethylene (PTFE) as the reference standard[1][2].

Uncertainty values take into account the uncertainties of the pressed PTFE reference standard. The 8/h spectral reflectance of the sample R_S at each wavelength λ was calculated from:

Eq. 1
$$R_S = \frac{M_S - M_H}{1 - M_H} \cdot R_R$$

where:
M_S is the instrument's relative measurement of the reflectance of the calibration item
M_H is the instrument's relative measurement of the reflectance of an open sample port and measures the effect of stray light overfilling the sample port
R_R is the 8/h spectral reflectance of the pressed PTFE reference standard.

[1] Wiedner V.R., and Hsia, J. J. "Reflection Properties of Pressed Polytetrafluoroethylene Powder", J.Opt.Soc.Am., Vol71, 1981, pp856-861

[2] Barnes, P.Y., Early, E.A., and Parr, A.C., "NIST Measuremetn Services: Spectral Reflectance," U.S. Dept. of Commerce, 1998.

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The instrument's relative measured reflectances, M_S and M_H are equivalent to:

Eq. 2

$$M_S = \frac{S_S}{S_R}$$

and

Eq. 3

$$M_H = \frac{S_H}{S_R}$$

where:
S_R is the signal with the reference standard in place
S_S is the signal with the calibration item in place
S_H is the signal with an open port

Therefore, the 8/h spectral reflectance of the calibration item R_S at each wavelength λ is equivalent to:

Eq. 4

$$R_S = \frac{S_S - S_H}{S_R - S_H} \cdot R_R$$

The final 8/h spectral reflectance is obtained by averaging the values from three scans.

Sources of uncertainty are:

- the 8/h spectral reflectance of the pressed PTFE reference standard
- nonuniformity across surfaces of the reference standard
- nonuniformity across the surface of the calibration item
- residual uncertainty in the correction for extraneous light
- nonlinearity of the instrument
- effect of wavelength error
- random noise in the measurements

The individual contributions to uncertainty are combined by adding in quadrature (root-sum-square). The quadrature sum is multiplied by a coverage factor (k) to generate the expanded uncertainty. The coverage factor is chosen to provide a confidence level of 95%. For effective degrees of freedom of thirty or greater, k = 2.0. For lower degrees of freedom a larger coverage factor is used representing the coverage factor necessary to provide a confidence level of 95% for a t-distribution with the corresponding degrees of freedom. Reference Table IV for the expended uncertainty of the calibration results.

Calibrated by:
Title: Optical Calibration Technician

Approved by:

General Information

1. The values in Table III apply only to the central 14 mm by 6 mm area of the items for measurement geometry 8/h.
2. The report of calibration may not be reproduced except in full without the written consent of this laboratory.
3. This report must not be used to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

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Table I
8°/Hemispherical Spectral Reflectance for SRT-MS-050 Multi-Step Reflectance Target

Wavelength (nm)	Reflectance	Wavelength (nm)	Reflectance
250	0.9647	1400	0.9892
300	0.9872	1450	0.9900
350	0.9931	1500	0.9901
400	0.9910	1550	0.9900
450	0.9902	1600	0.9903
500	0.9911	1650	0.9891
550	0.9904	1700	0.9878
600	0.9907	1750	0.9877
650	0.9908	1800	0.9879
700	0.9905	1850	0.9832
750	0.9905	1900	0.9823
800	0.9894	1950	0.9807
850	0.9898	2000	0.9762
900	0.9921	2050	0.9652
950	0.9923	2100	0.9580
1000	0.9924	2150	0.9543
1050	0.9924	2200	0.9695
1100	0.9923	2250	0.9701
1150	0.9923	2300	0.9637
1200	0.9912	2350	0.9547
1250	0.9912	2400	0.9505
1300	0.9902	2450	0.9479
1350	0.9898	2500	0.9512

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Table II
8°/Hemispherical Spectral Reflectance for SRT-MS-050 Multi-Step Reflectance Target

Wavelength (nm)	Reflectance	Wavelength (nm)	Reflectance
250	0.4405	1400	0.5651
300	0.4336	1450	0.5699
350	0.4339	1500	0.5744
400	0.4391	1550	0.5773
450	0.4466	1600	0.5823
500	0.4555	1650	0.5861
550	0.4640	1700	0.5889
600	0.4721	1750	0.5917
650	0.4801	1800	0.5955
700	0.4877	1850	0.5982
750	0.4951	1900	0.6015
800	0.5022	1950	0.6045
850	0.5090	2000	0.6088
900	0.5167	2050	0.6167
950	0.5235	2100	0.6202
1000	0.5286	2150	0.6219
1050	0.5334	2200	0.6261
1100	0.5389	2250	0.6258
1150	0.5440	2300	0.6230
1200	0.5478	2350	0.6398
1250	0.5533	2400	0.6462
1300	0.5569	2450	0.6355
1350	0.5607	2500	0.6411

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Table III
8°/Hemispherical Spectral Reflectance for SRT-MS-050 Multi-Step Reflectance Target

Wavelength (nm)	Reflectance	Wavelength (nm)	Reflectance
250	0.2178	1400	0.3194
300	0.2092	1450	0.3241
350	0.2069	1500	0.3285
400	0.2090	1550	0.3329
450	0.2142	1600	0.3372
500	0.2199	1650	0.3412
550	0.2260	1700	0.3453
600	0.2325	1750	0.3489
650	0.2388	1800	0.3527
700	0.2448	1850	0.3558
750	0.2509	1900	0.3594
800	0.2566	1950	0.3628
850	0.2643	2000	0.3679
900	0.2691	2050	0.3723
950	0.2748	2100	0.3766
1000	0.2803	2150	0.3807
1050	0.2857	2200	0.3810
1100	0.2910	2250	0.3836
1150	0.2961	2300	0.3872
1200	0.3009	2350	0.3919
1250	0.3058	2400	0.3945
1300	0.3104	2450	0.3962
1350	0.3149	2500	0.4014

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Table IV
8º/Hemispherical Spectral Reflectance for SRT-MS-050 Multi-Step Reflectance Target

Wavelength (nm)	Reflectance factor	Wavelength (nm)	Reflectance
250	0.1231	1400	0.1471
300	0.1149	1450	0.1494
350	0.1123	1500	0.1516
400	0.1111	1550	0.1537
450	0.1113	1600	0.1559
500	0.1124	1650	0.1580
550	0.1136	1700	0.1601
600	0.1149	1750	0.1622
650	0.1167	1800	0.1642
700	0.1186	1850	0.1660
750	0.1205	1900	0.1682
800	0.1223	1950	0.1701
850	0.1252	2000	0.1727
900	0.1266	2050	0.1755
950	0.1285	2100	0.1780
1000	0.1305	2150	0.1812
1050	0.1326	2200	0.1814
1100	0.1347	2250	0.1828
1150	0.1368	2300	0.1852
1200	0.1388	2350	0.1882
1250	0.1409	2400	0.1904
1300	0.1429	2450	0.1936
1350	0.1450	2500	0.2060

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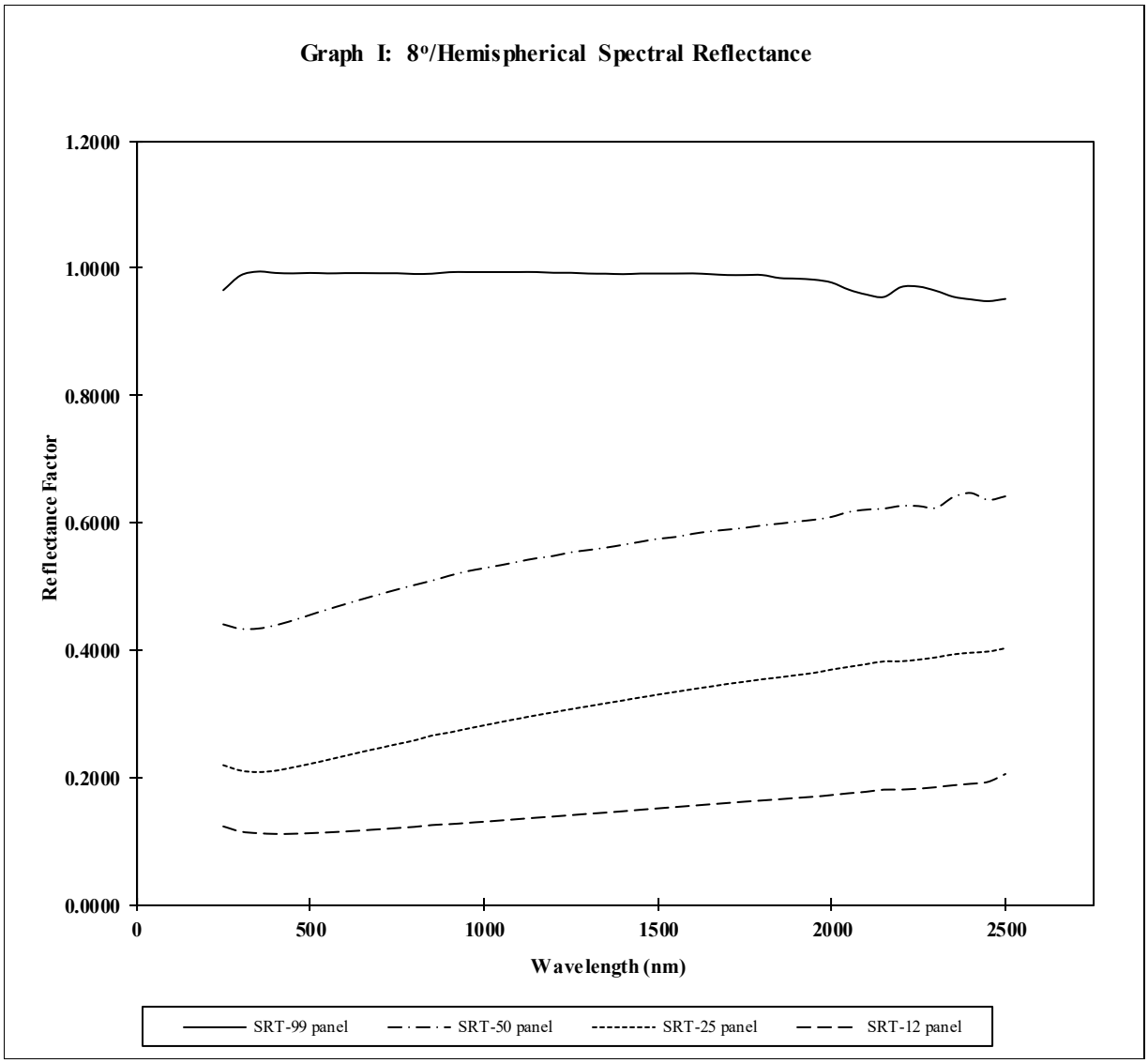

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Table V Uncertainty contributions and expanded uncertainty (k=2) of the 8/h spectral reflectance for the measured sample

CALIBRATION AND MEASUREMENT CAPABILITIES (CMC) Notes 1,2

Measured Parameter or Device Calibrated	Range	Uncertainty (*k*=2) Note 3,5	Remarks
OPTICAL RADIATION			
Photometric (20/O02)			
Relative Reflectance at Wavelength Shown Below:			Relative reflectance is a dimensionless quantity
250 nm to 600 nm	up to 0.02	0.0016	
	> 0.02 to 0.05	0.0029	
	> 0.05 to 0.10	0.012	
	> 0.10 to 0.20	0.012	
	> 0.20 to 0.50	0.0054	
	> 0.50 to 0.80	0.0054	
	> 0.80 to 0.99	0.0053	
601 nm to 1500 nm	up to 0.02	0.0017	
	> 0.02 to 0.05	0.0022	
	> 0.05 to 0.10	0.0025	
	> 0.10 to 0.20	0.0052	
	> 0.20 to 0.50	0.0064	
	> 0.50 to 0.80	0.0064	
	> 0.80 to 0.99	0.0049	
1501 nm to 2200 nm	up to 0.02	0.0090	
	> 0.02 to 0.05	0.0090	
	> 0.05 to 0.10	0.015	
	> 0.10 to 0.20	0.015	
	> 0.20 to 0.50	0.0099	
	> 0.50 to 0.80	0.0083	
	> 0.80 to 0.99	0.0088	
2201 nm to 2500 nm	up to 0.02	0.054	
	> 0.02 to 0.05	0.054	
	> 0.05 to 0.10	0.043	
	> 0.10 to 0.20	0.043	
	> 0.20 to 0.50	0.035	
	> 0.50 to 0.80	0.028	
	> 0.80 to 0.99	0.032	
END			

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Notes
Note 1: A Calibration and Measurement Capability (CMC) is a description of the best result of a calibration or measurement (result with the smallest uncertainty of measurement) that is available to the laboratory's customers under normal conditions, when performing more or less routine calibrations of nearly ideal measurement standards or instruments. The CMC is described in the laboratory's scope of accreditation by: the measurement parameter/device being calibrated, the measurement range, the uncertainty associated with that range (see note 3), and remarks on additional parameters, if applicable. **Note 2:** Calibration and Measurement Capabilities are traceable to the national measurement standards of the U.S. or to the national measurement standards of other countries and are thus traceable to the internationally accepted representation of the appropriate SI (Système International) unit. **Note 3:** The uncertainty associated with a measurement in a CMC is an expanded uncertainty using a coverage factor, $k = 2$, with a level of confidence of approximately 95 %. Units for the measurand and its uncertainty are to match. Exceptions to this occur when marketplace practice employs mixed units, such as when the artifact to be measured is labeled in non-SI units and the uncertainty is given in SI units (Example: 5 lb weight with uncertainty given in mg). **Note 3a:** The uncertainty of a specific calibration by the laboratory may be greater than the uncertainty in the CMC due to the condition and behavior of the customer's device and specific circumstances of the calibration. The uncertainties quoted do not include possible effects on the calibrated device of transportation, long term stability, or intended use. **Note 3b:** As the CMC represents the best measurement results achievable under normal conditions, the accredited calibration laboratory shall not report smaller uncertainty of measurement than that given in a CMC for calibrations or measurements covered by that CMC. **Note 3c:** As described in Note 1, CMCs cover calibrations and measurements that are available to the laboratory's customers under *normal conditions.* However, the laboratory may have the capability to offer special tests, employing special conditions, which yield calibration or measurement results with lower uncertainties. Such special tests are not covered by the CMCs and are outside the laboratory's scope of accreditation. In this case, NVLAP requirements for the labeling, on calibration reports, of results outside the laboratory's scope of accreditation apply. These requirements are set out in Annex A.1.h. of NIST Handbook 150, Procedures and General Requirements. **Note 4:** Uncertainties associated with field service calibration may be greater as they incorporate on-site environmental contributions, transportation effects, or other factors that affect the measurements. (This note applies only if marked in the body of the scope.) **Note 5:** Values listed with percent (%) are percent of reading or generated value unless otherwise noted. **Note 6:** NVLAP accreditation is the formal recognition of specific calibration capabilities. Neither NVLAP nor NIST guarantee the accuracy of individual calibrations made by accredited laboratories. **Note 7:** See NIST Handbook 150 for further explanation of these notes.